

Systems of Objects

(Forces 2)

System- a combination of two or more masses

$$\sum F_{\text{net ext}} = m_{(\text{sys})} a_{\text{system}}$$

Internal and External forces

Internal forces - forces exerted by objects inside the defined system.

External forces - forces exerted by object outside the defined system.

Solving problems for a system

Method 1-

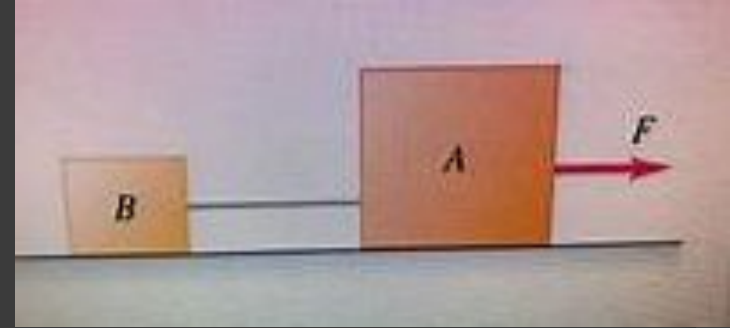
- Draw a FBD for each object in the system.
- Write an equation of motion for each object.
- Solve the equations for the unknown.

Solving problems for a system

Method 2

- Draw an FBD for the entire system
- Draw an FBD for each object in the system
- In the FBD for each object, identify the internal and external forces
- Exclude the internal forces and any external force pairs that cancel
- Get the net of the external forces
- ⦿ If the system is at rest or moving with constant velocity, set the net external force = 0.
- ⦿ if the system is accelerating, set the net external force = ma .
- ⦿ Solve from there.

Sample problem



A 5.0 kg object (m_1) is connected to a 10.0 kg object (m_2) by a string. If a pulling force F of 20 N is applied to the 5.0 kg object as shown,

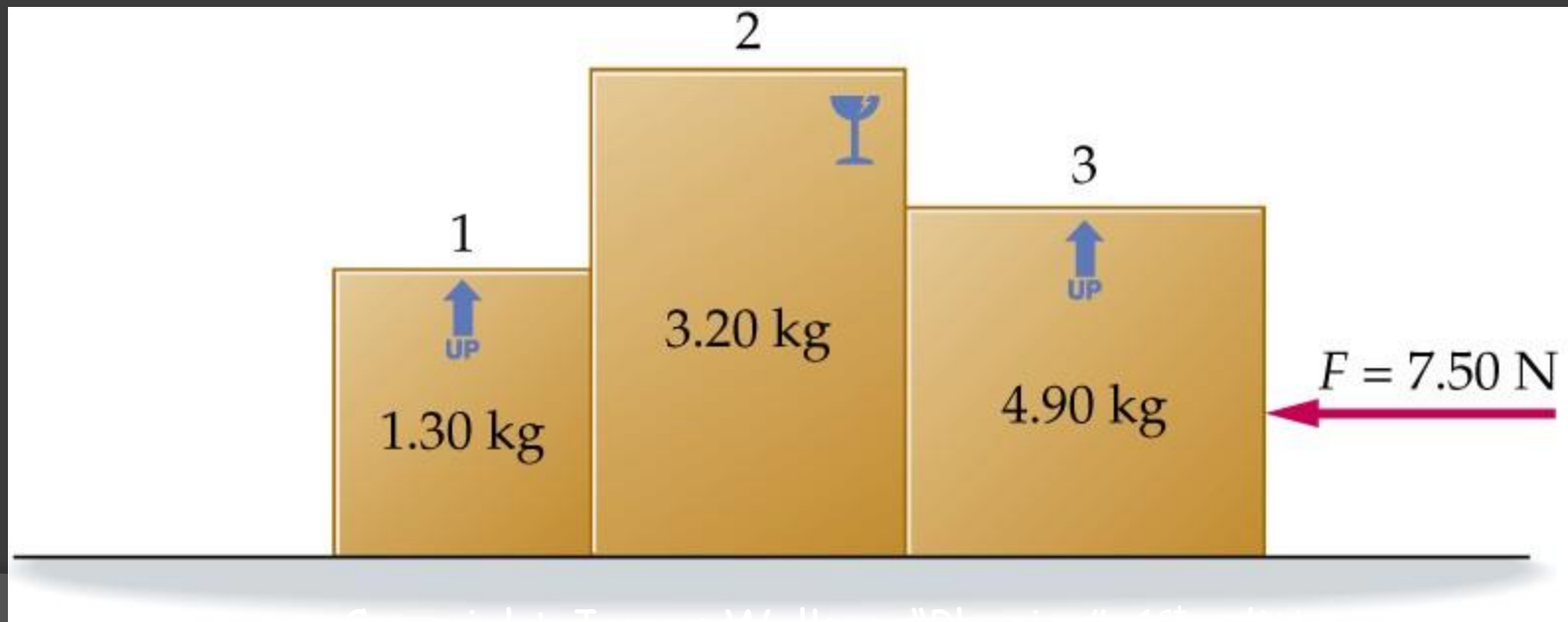
A) what is the acceleration of the system?

B) what is the tension in the string connecting the objects?

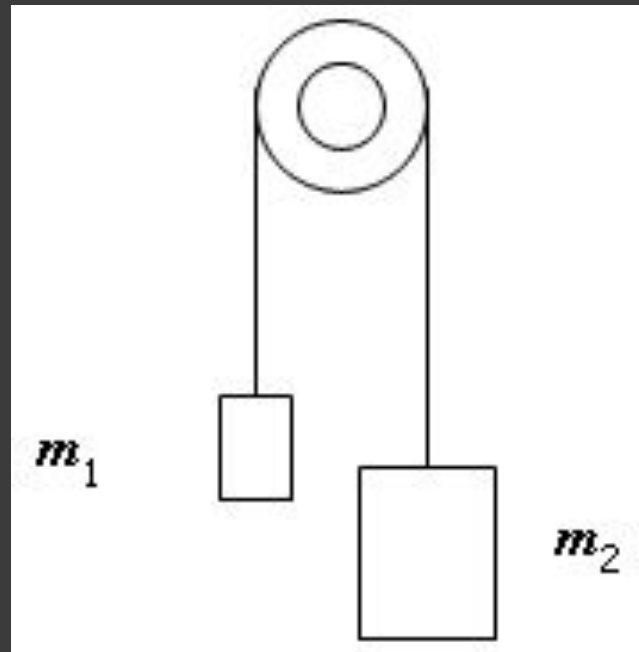
(Assume a frictionless surface.)

Sample Problem

A force of magnitude 7.50 N pushes three boxes with masses $m_1 = 1.30$ kg, $m_2 = 3.20$ kg, and $m_3 = 4.90$ kg as shown. Find the contact force between (a) boxes 1 and 2 and (b) between boxes 2 and 3.



An Atwood's machine has $m_1 = 1\text{kg}$, $m_2 = 2\text{kg}$, hung from an ideal pulley. What is the acceleration of the masses? Calculate the tension in the string attached to each mass.



74. ■ Two blocks are connected by a string as in Figure P5.74. What is the upper block's acceleration if the coefficient of kinetic friction between the block and the table is 0.20?

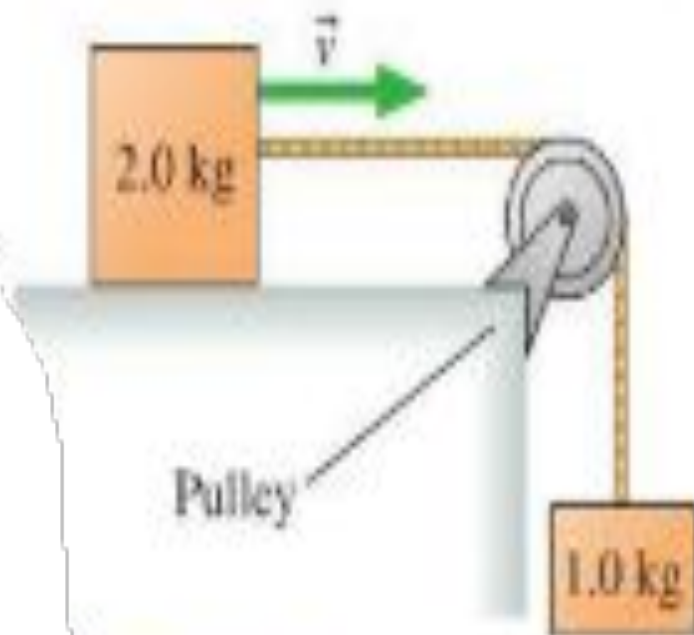


FIGURE P5.74